

SANTIAGO HOPKINS

Engineering Portfolio

*find out more about me and my
professional experience on my site!*

www.santiago.pw

E: SANTIAGOHOPKINS@BERKELEY.EDU

C: 818-383-5442

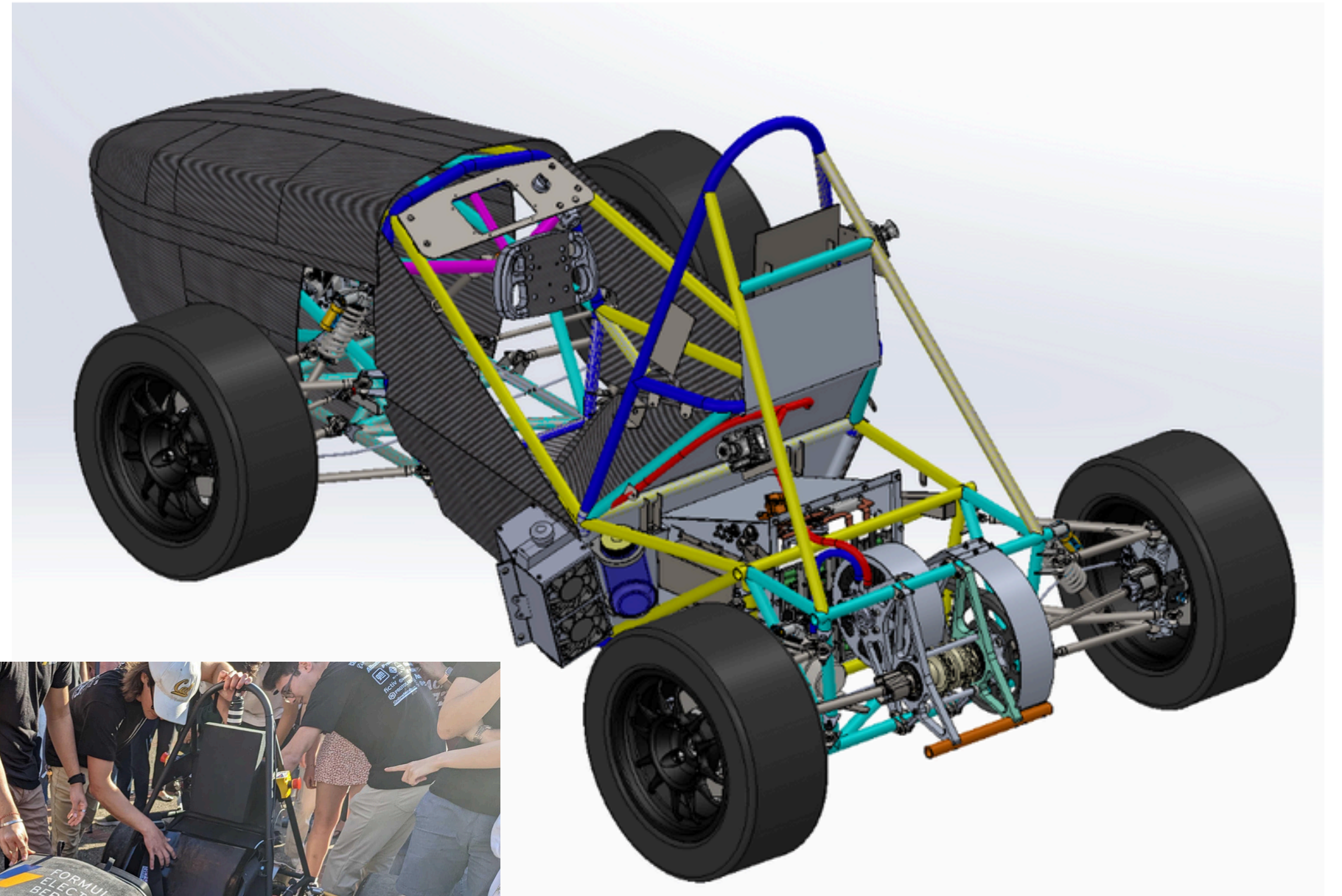


FSAE ELECTRIC DRIVERLESS

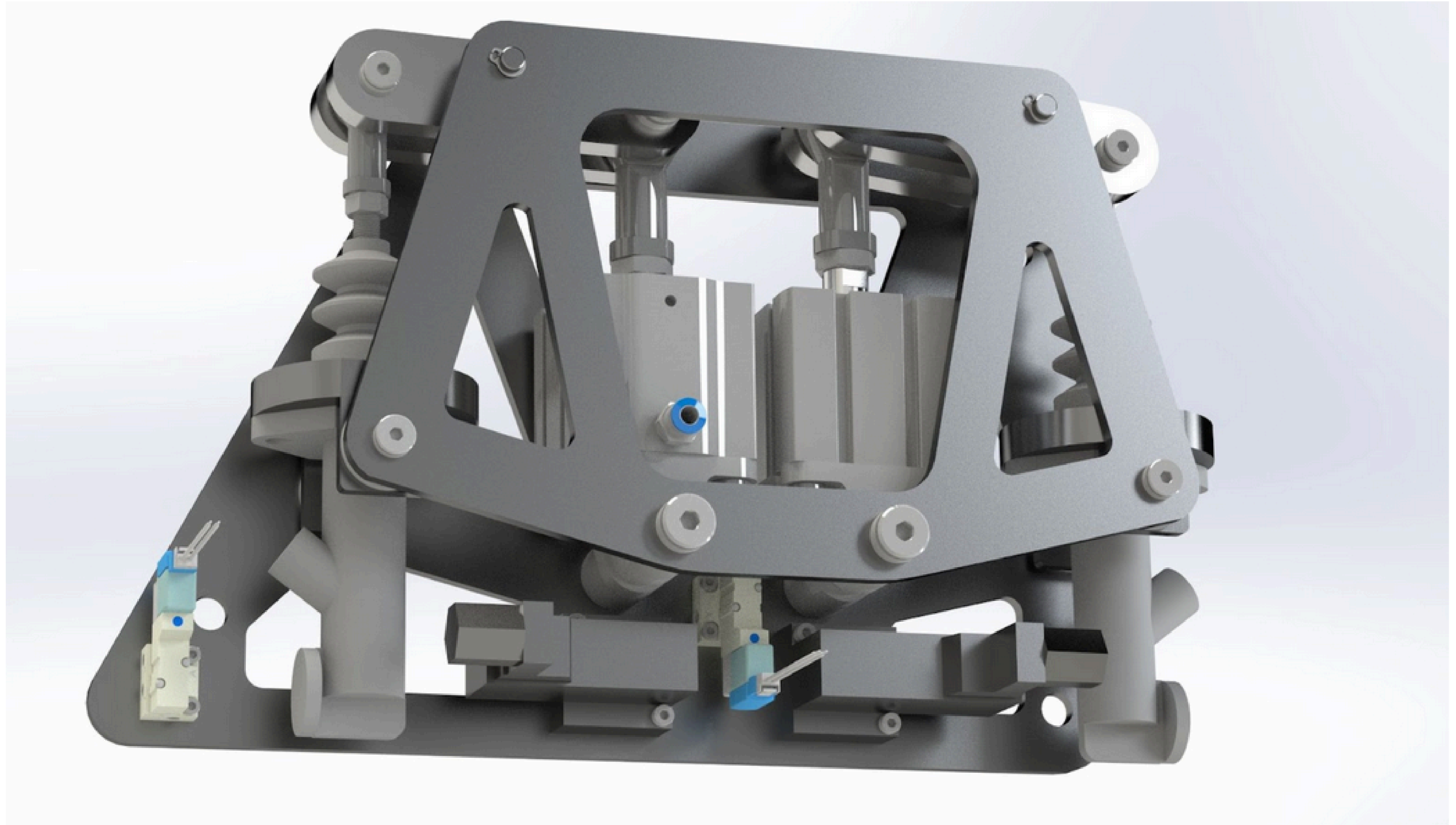


Founding Member
Driverless Mechanical Design

Lead
Autonomous Mechanical Systems



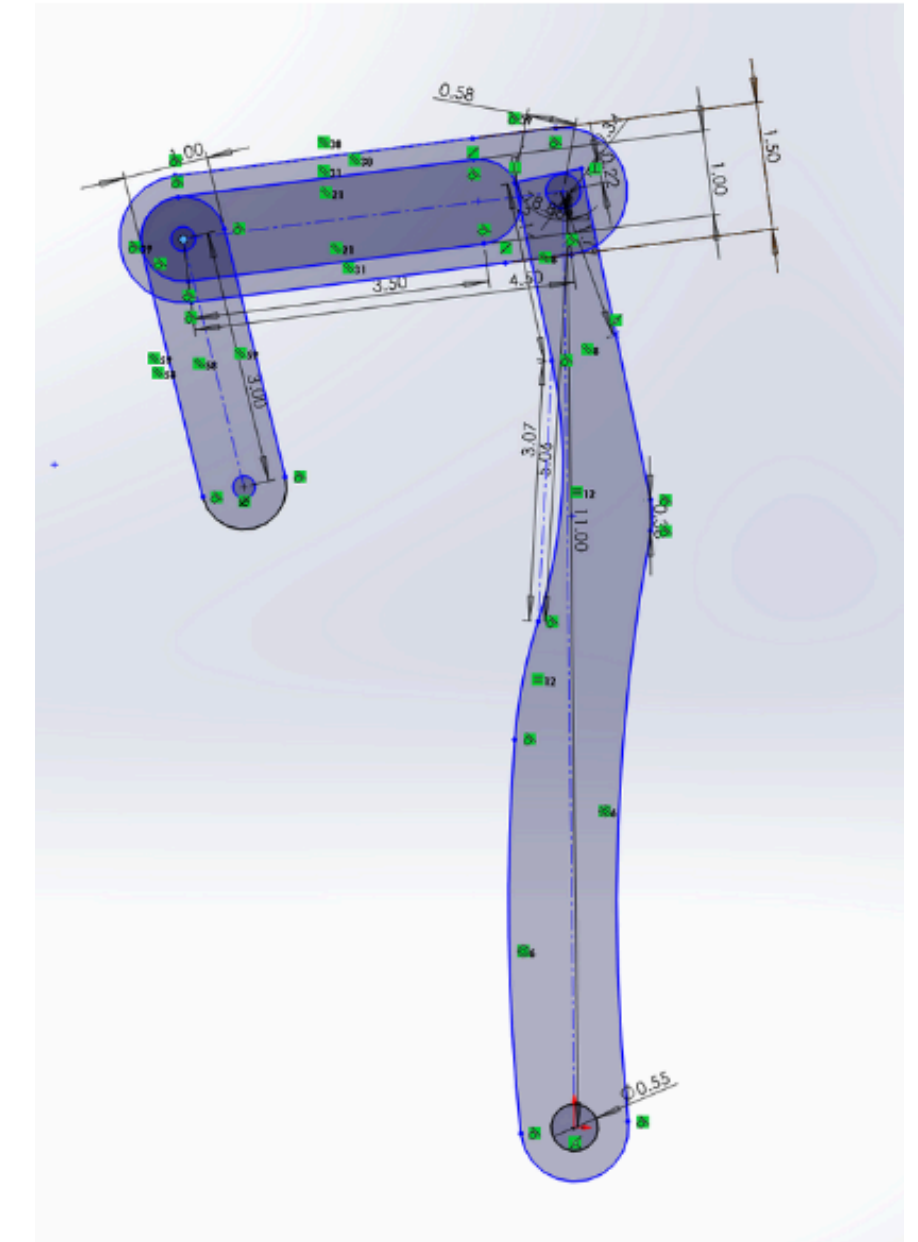
Autonomous Braking System



Autonomous Braking System

Details

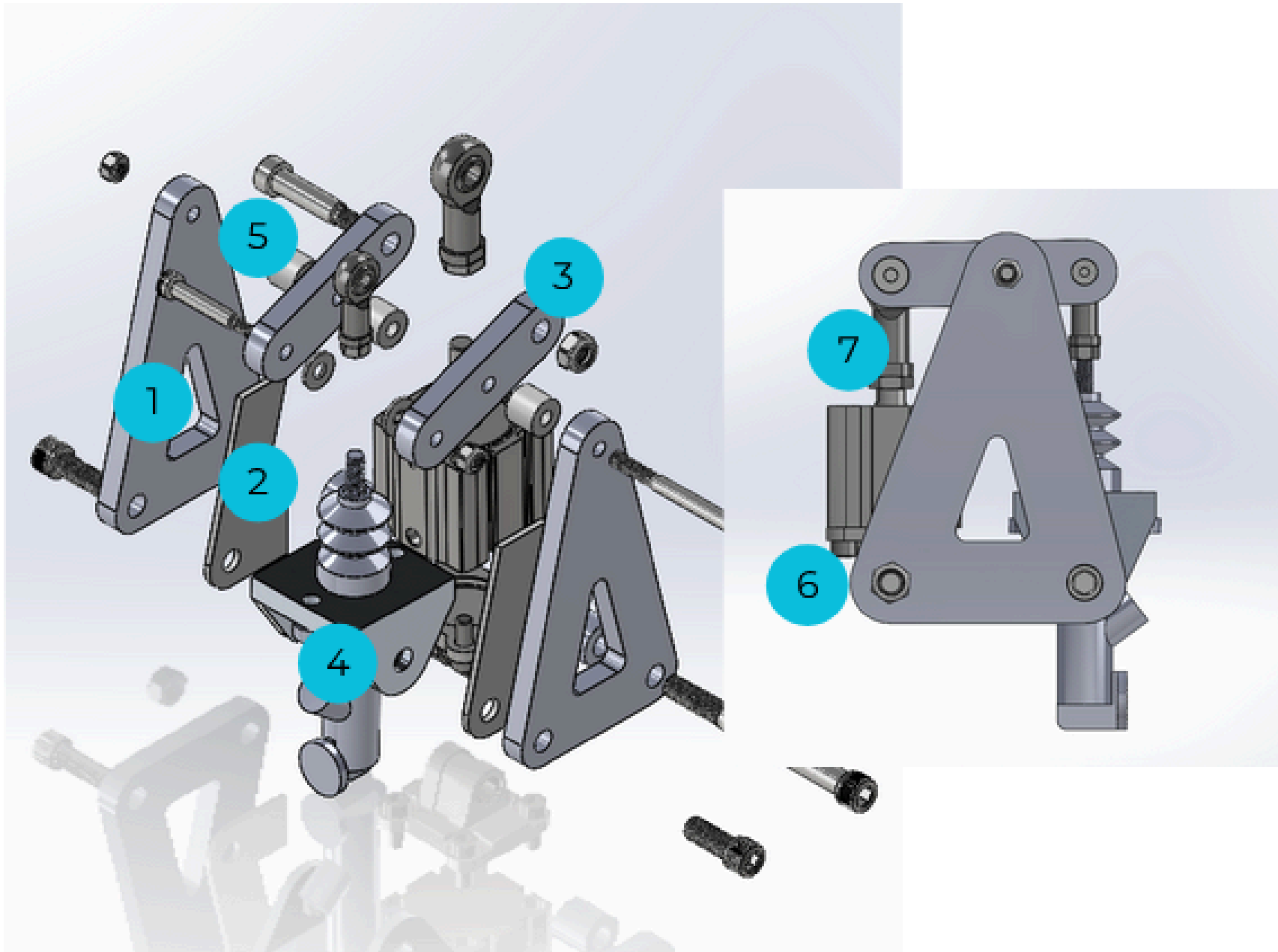
- Designed to fulfill '*passive mechanical fail-safe*' requirement
 - i.e. if power went out, our vehicle would actuate the brakes without any extra effort
- Modular, auxiliary system allows for test driver to retain manual braking input
- Pneumatic cylinders were able to apply upwards of 1000kN to pressurize brake line
- Response time of ~180ms
- Firmware and Other electrical fail-safes developed as well!



'Robot Foot': Early concept of ABS - Scrapped due to common cause failure concerns and space constraints

Autonomous Braking System

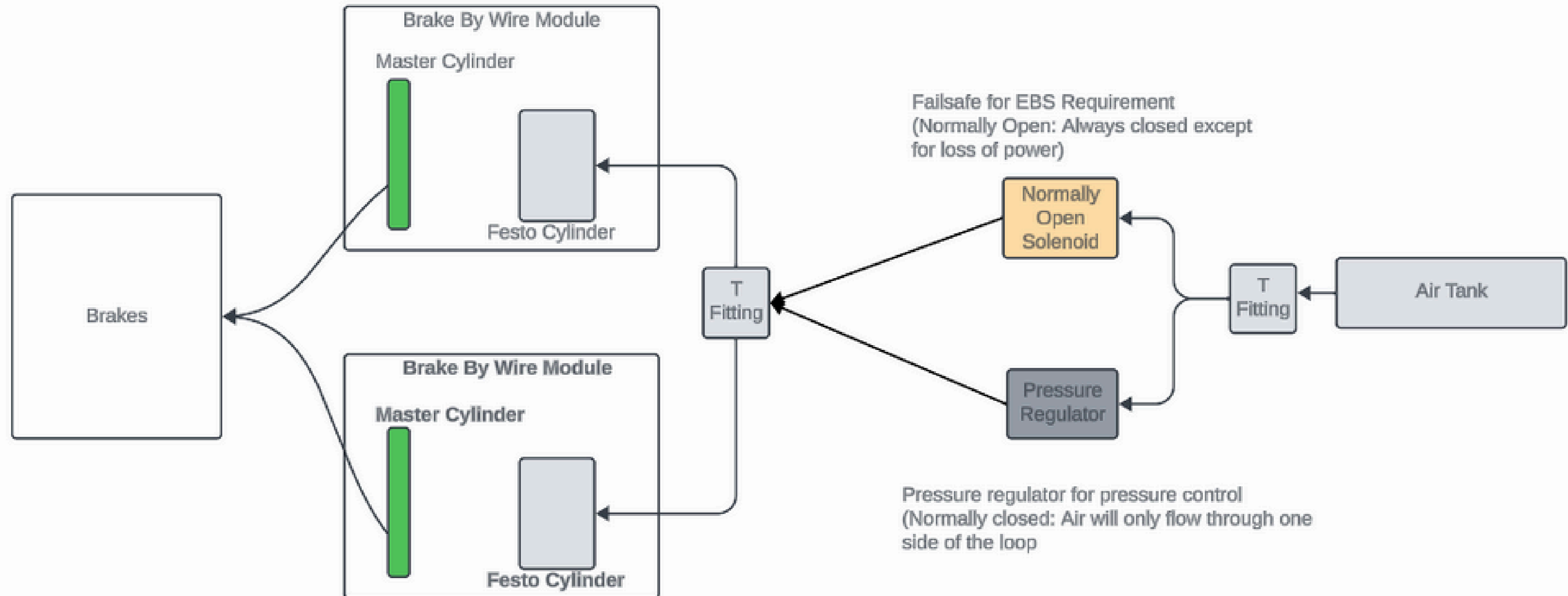
Mechanical Design



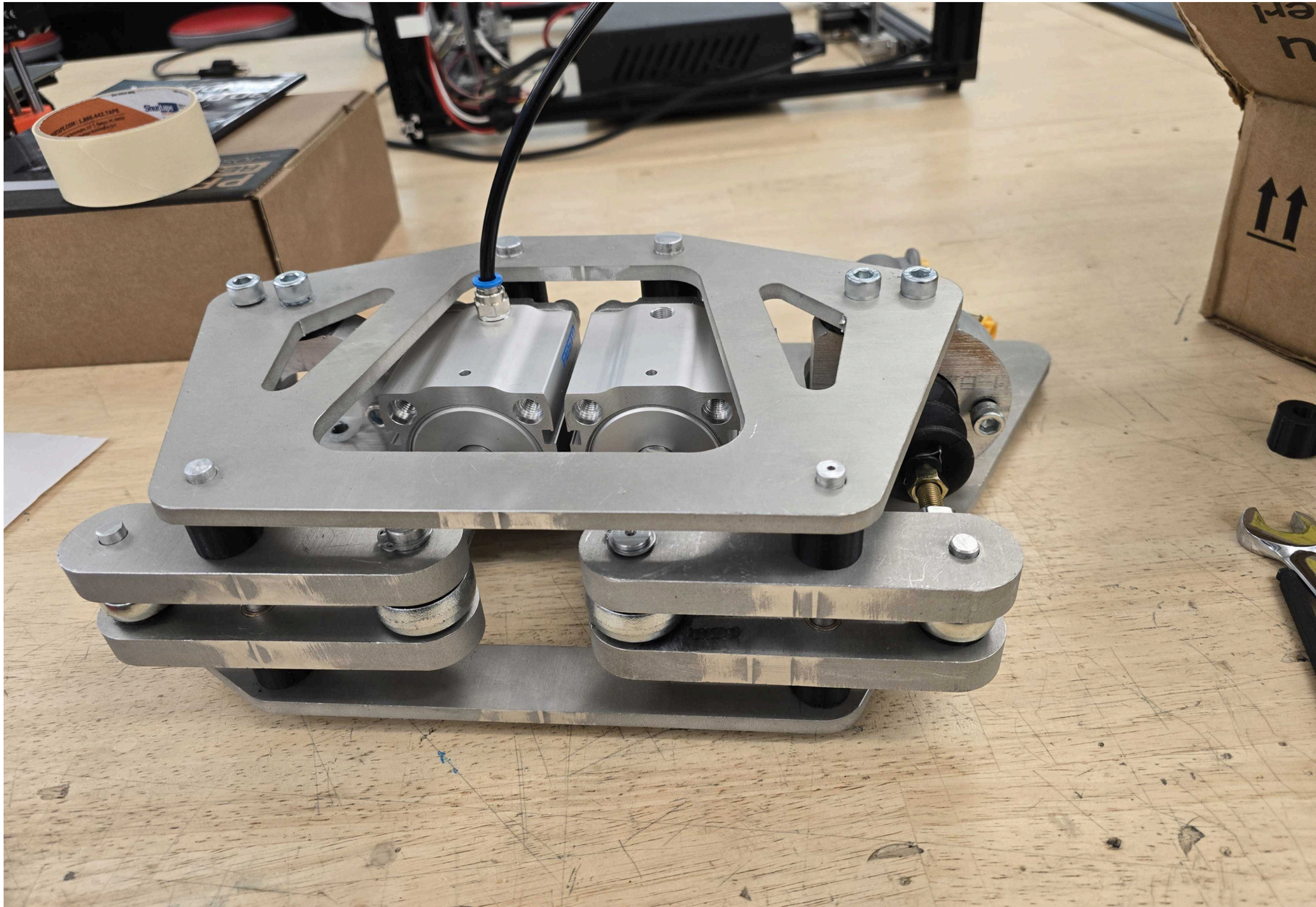
- 1 Triangular Panels
- 2 Plastic Wear Pads
- 3 Lever
- 4 Pivot Mount
- 5 Plastic Cylindrical Spacers
- 6 COTS Pivot Mount
- 7 COTS Rod Eye

Autonomous Braking System

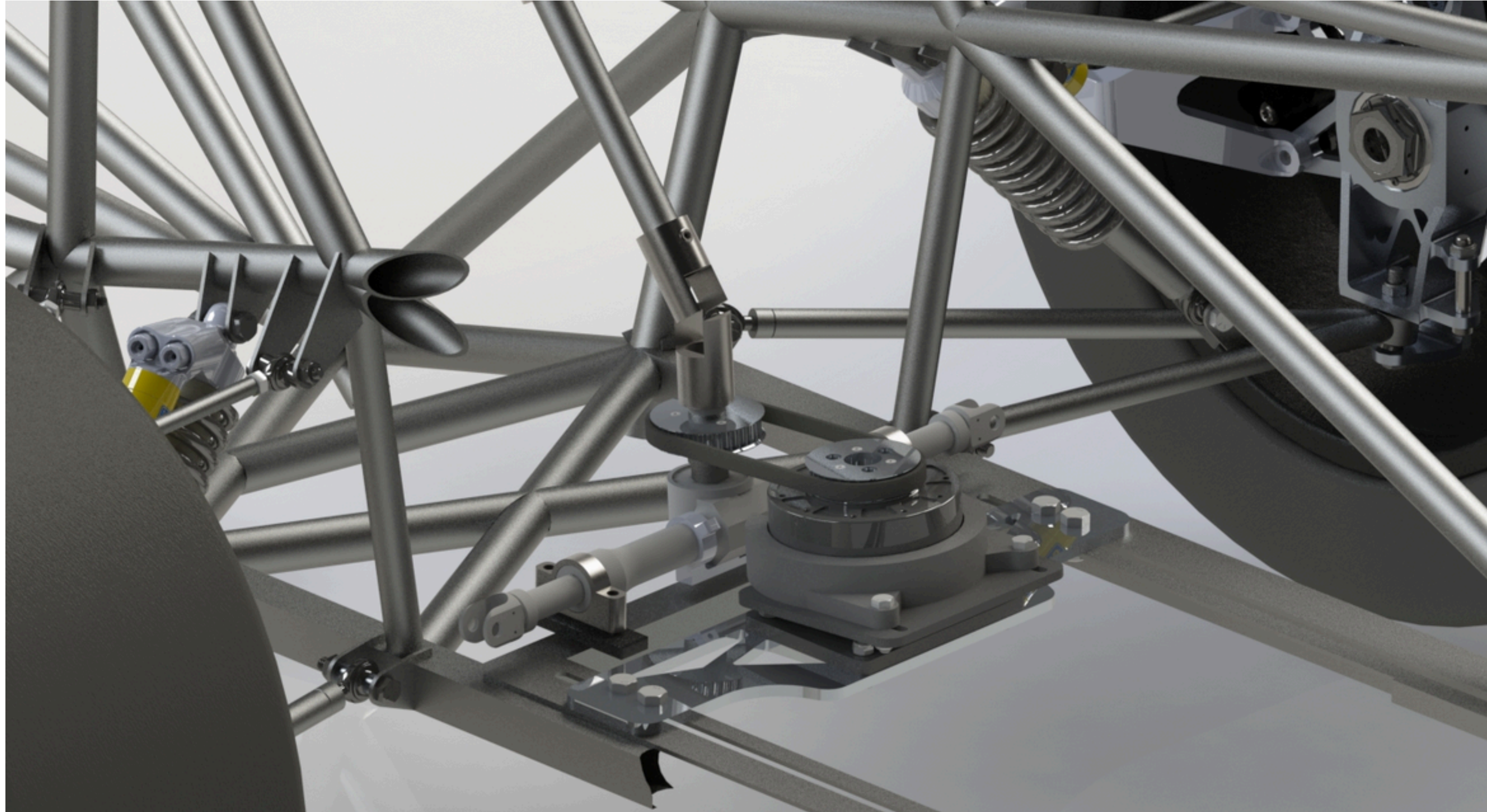
Pneumatic Circuit



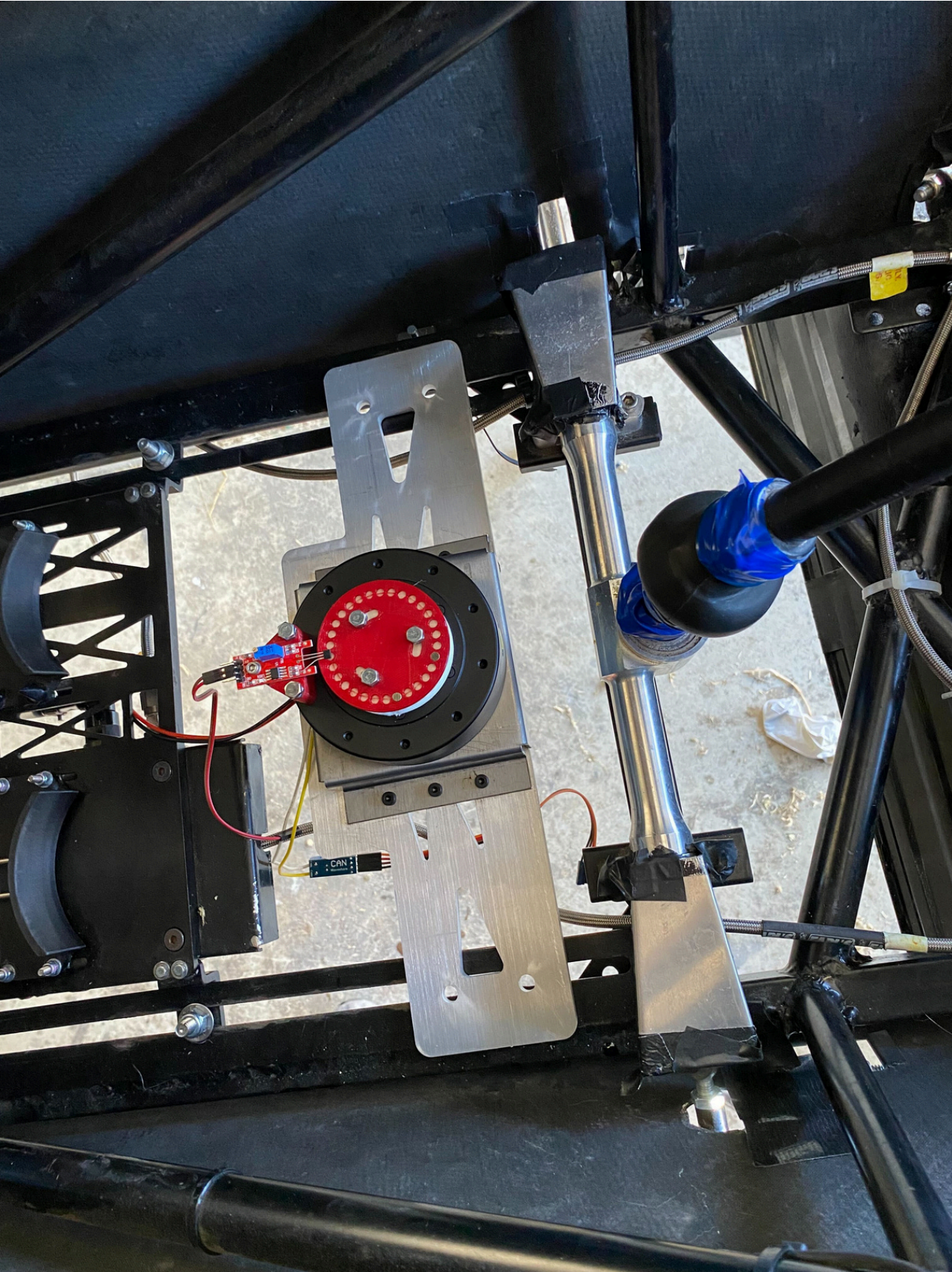
Autonomous Braking System



Autonomous Steering System



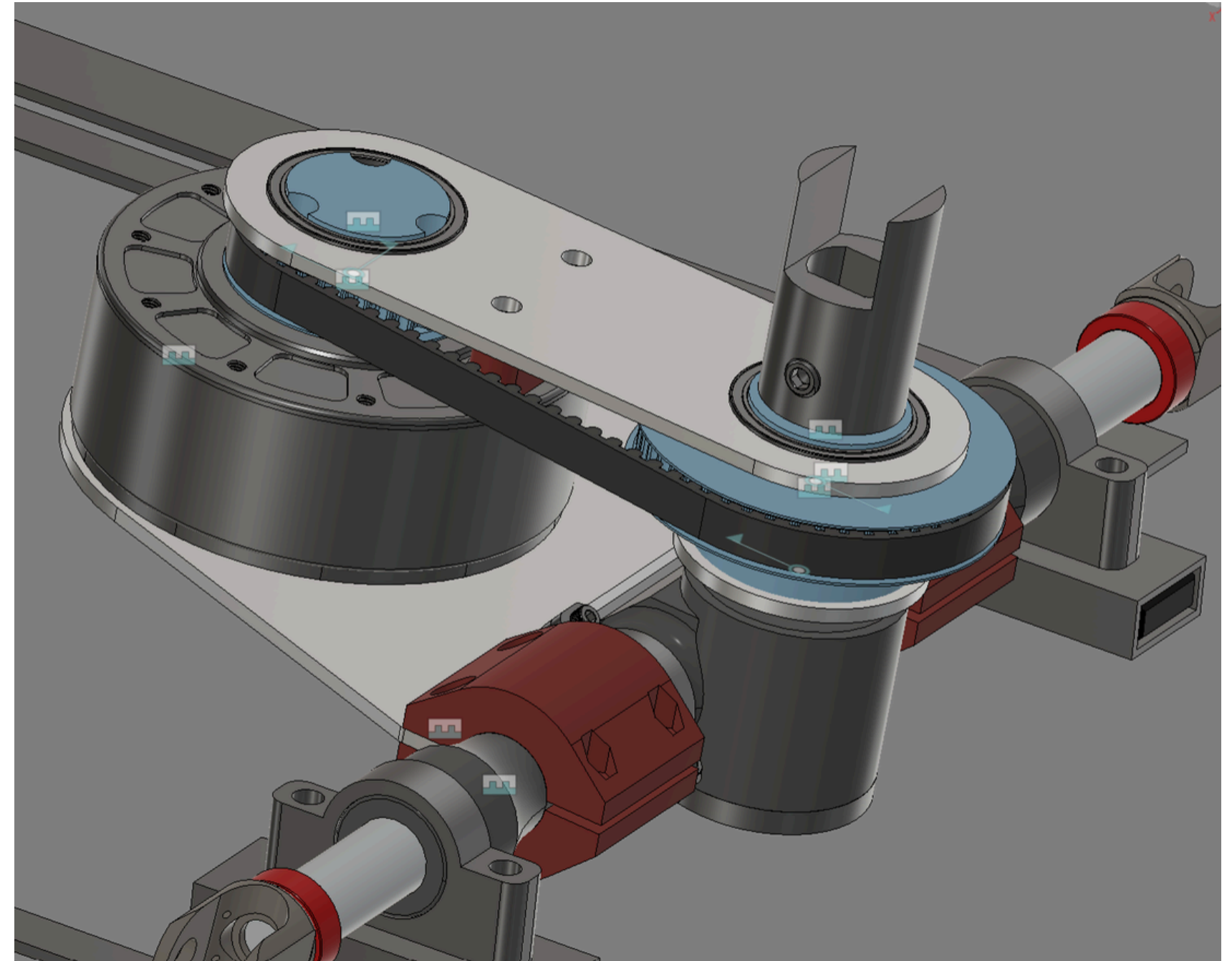
Autonomous Steering System



Autonomous Steering System

Details

- Designed to fulfill template test requirement
 - i.e. cockpit must be effectively empty of components (minus the steering column)
 - Built to be low-profile (level with steering rack)
- RMD x8 Pro 1:6 Planetary Gearbox
 - CAN communication
 - Robust built-in controls
- Belt Drive reduction of 1:1.5 allows for ~9Nm of torque
- HTD5m x 9mm belt with double flanged timing pulleys

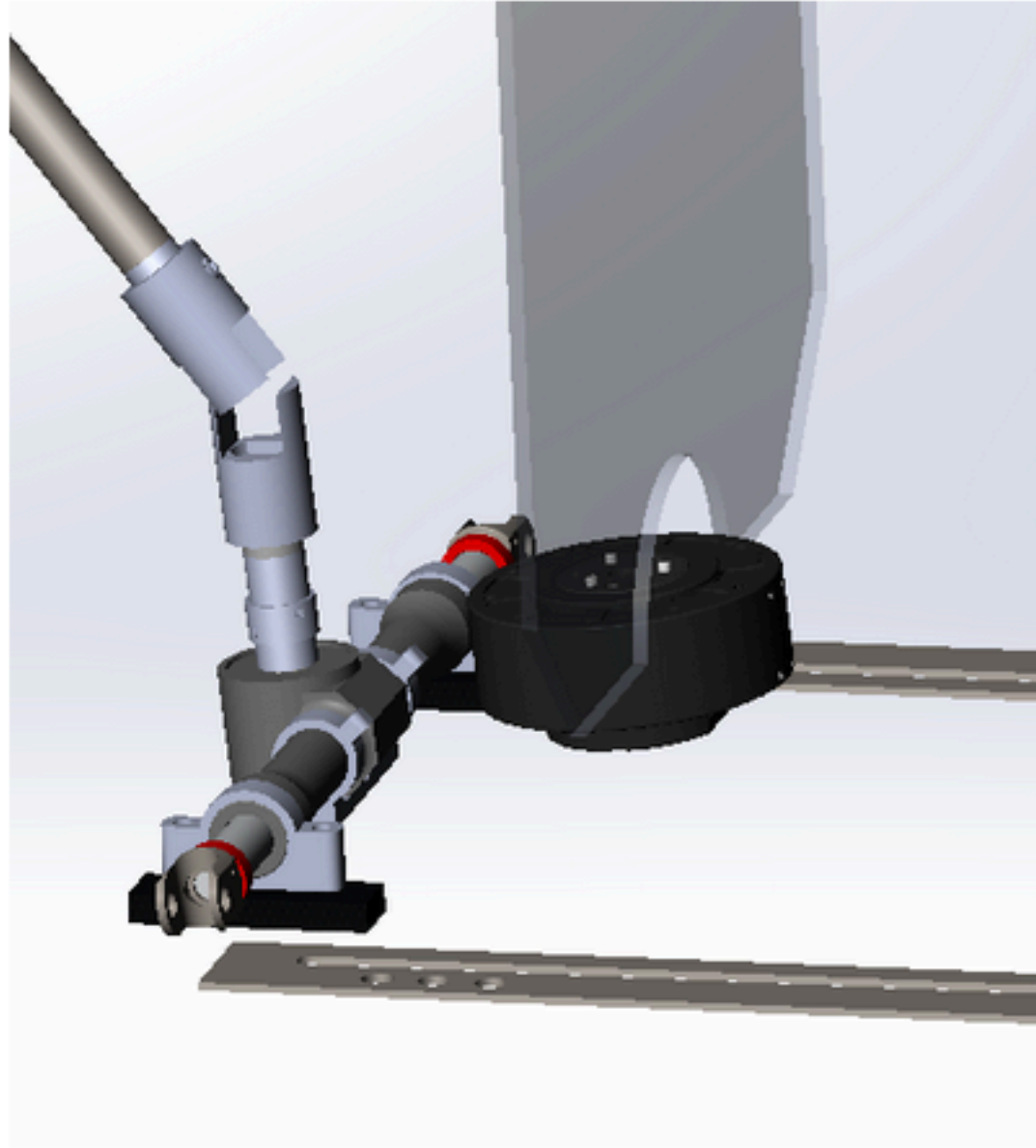


*Latest Autonomous Steering System
mount, designed to account for
displacement in steering column*

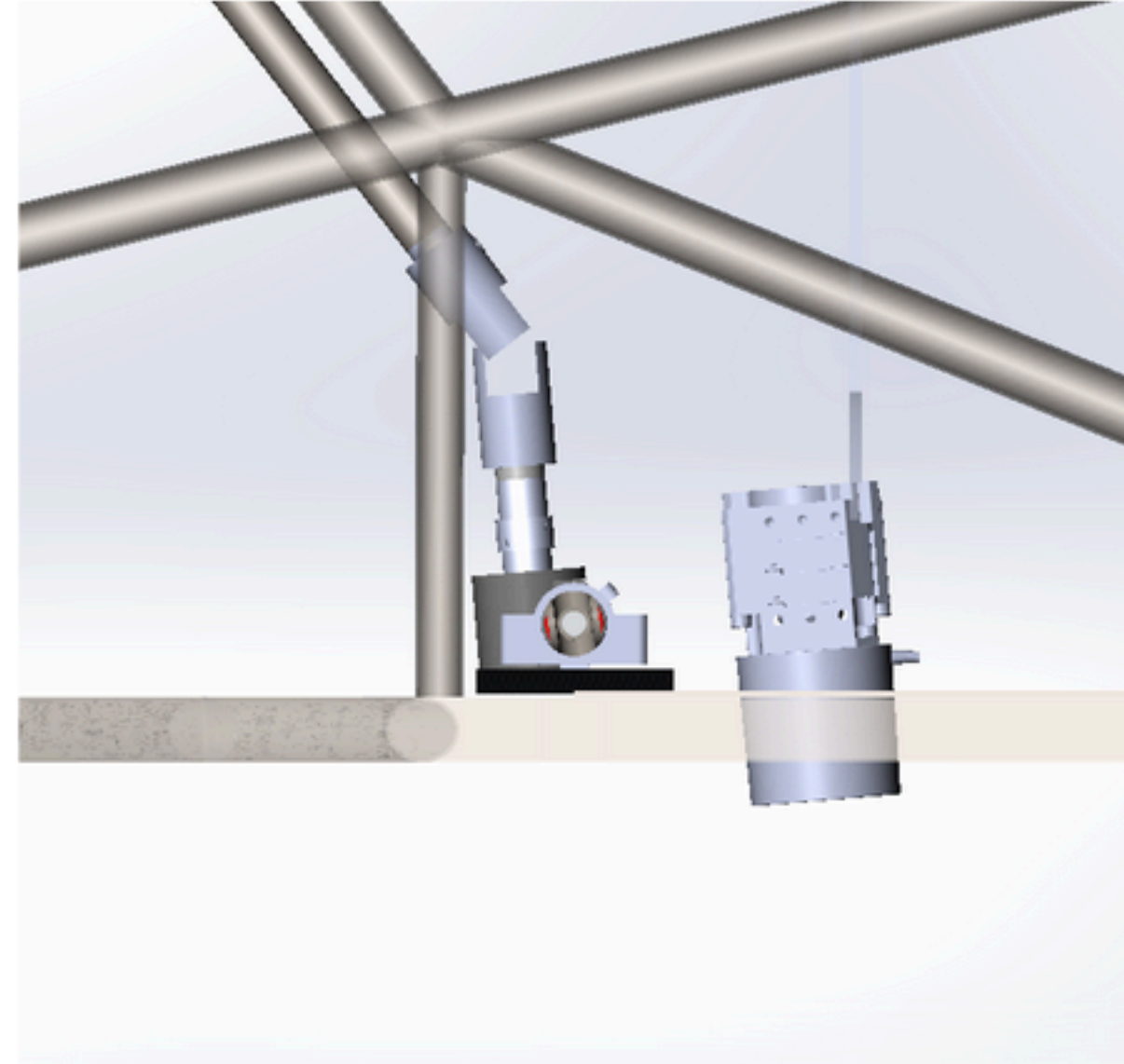
Autonomous Steering System

Other Concepts

RMDx8

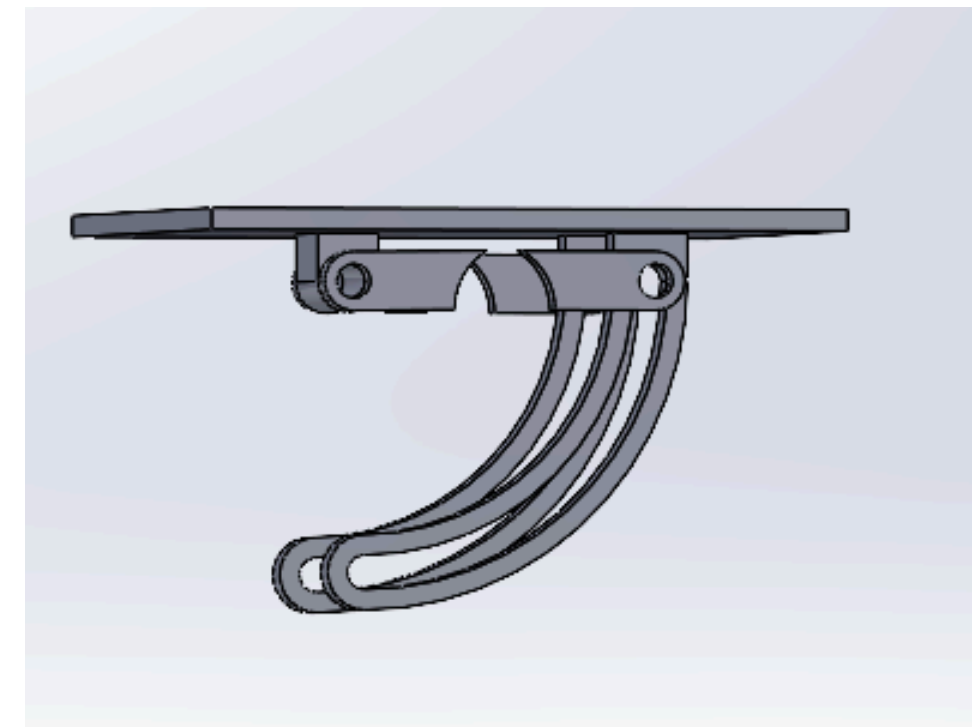
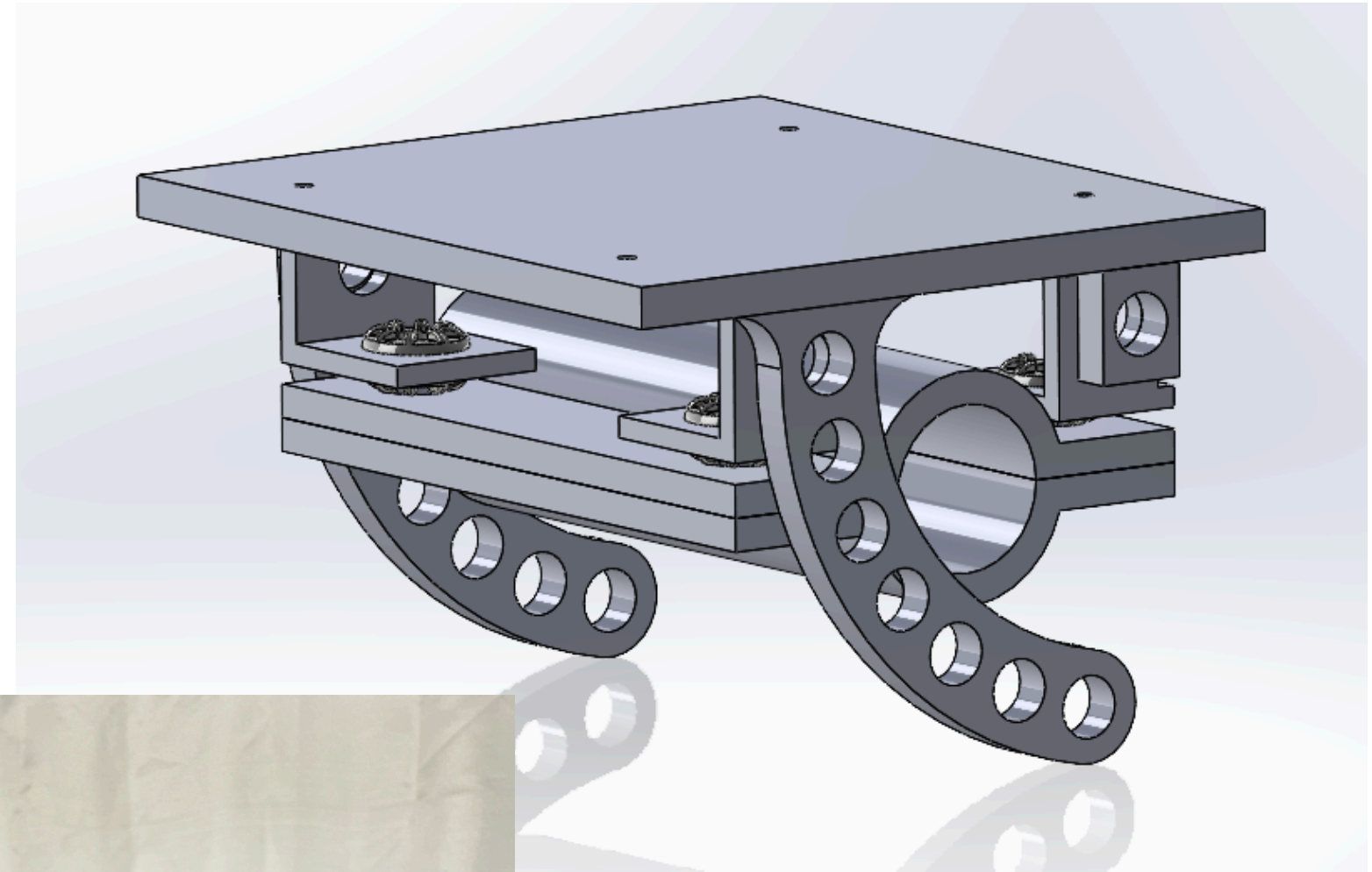


NeoRev



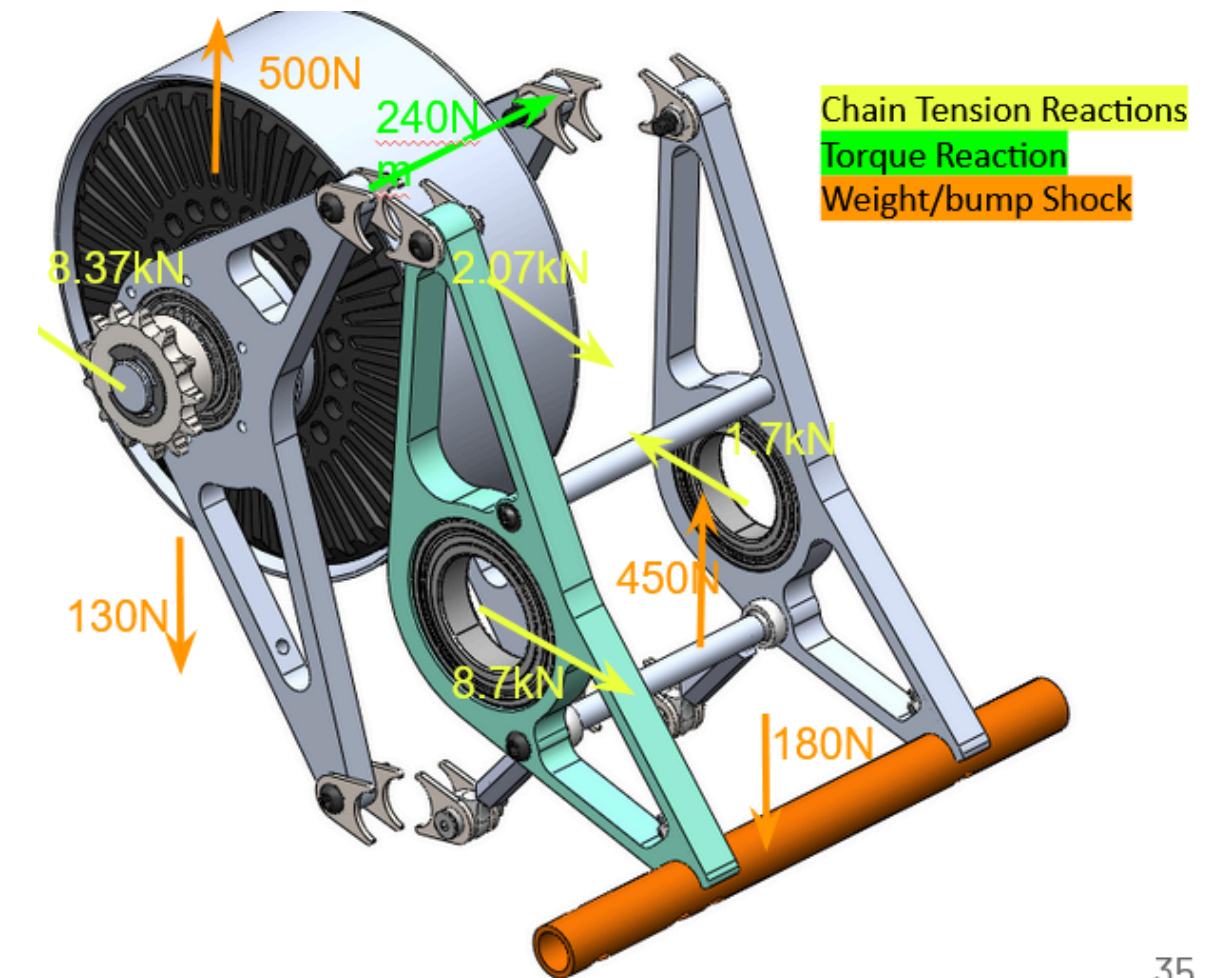
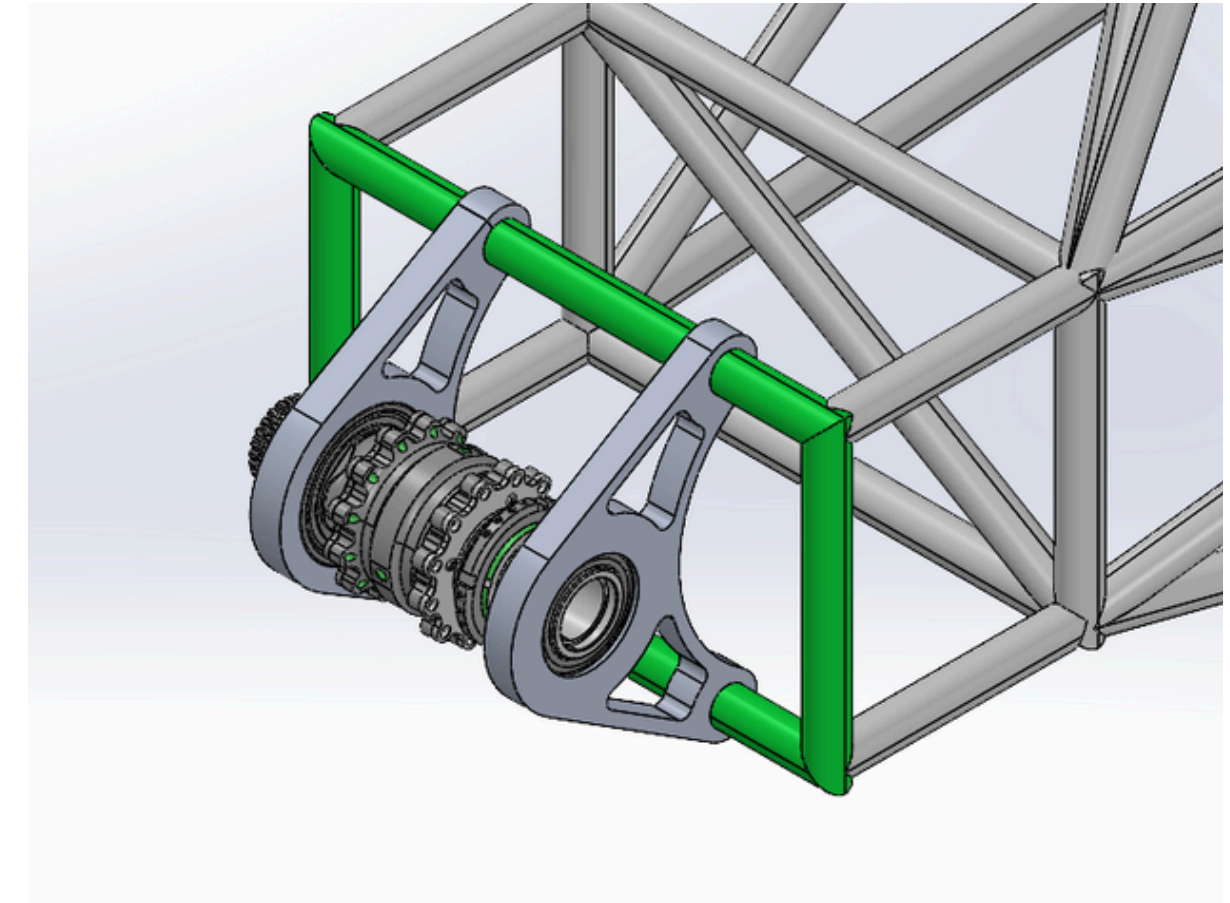
LiDAR Mount

- Designed and Fabricated LiDAR Mount
- Rule Compliant
- ~35° Range of Motion
- Rapid Prototyping via 3D Printing
- High Infill PLA
- Anti-Vibration
 - Sandwich Mounts -> Anti-Vibration Grommets
- Collected data rapidly and effectively for software pipeline training
- -> Aluminum carriage mounted via welded steel tabs.



Differential Mount

- Drafted early concepts for Differential Mount
- Minimized weight, while retaining sturdiness via FEA and Topology Studies
- Features:
 - Turnbuckle Tensioning
 - Mounts to mounting tabs for easier manufacturing and adjustment



WORLD SURF LEAGUE

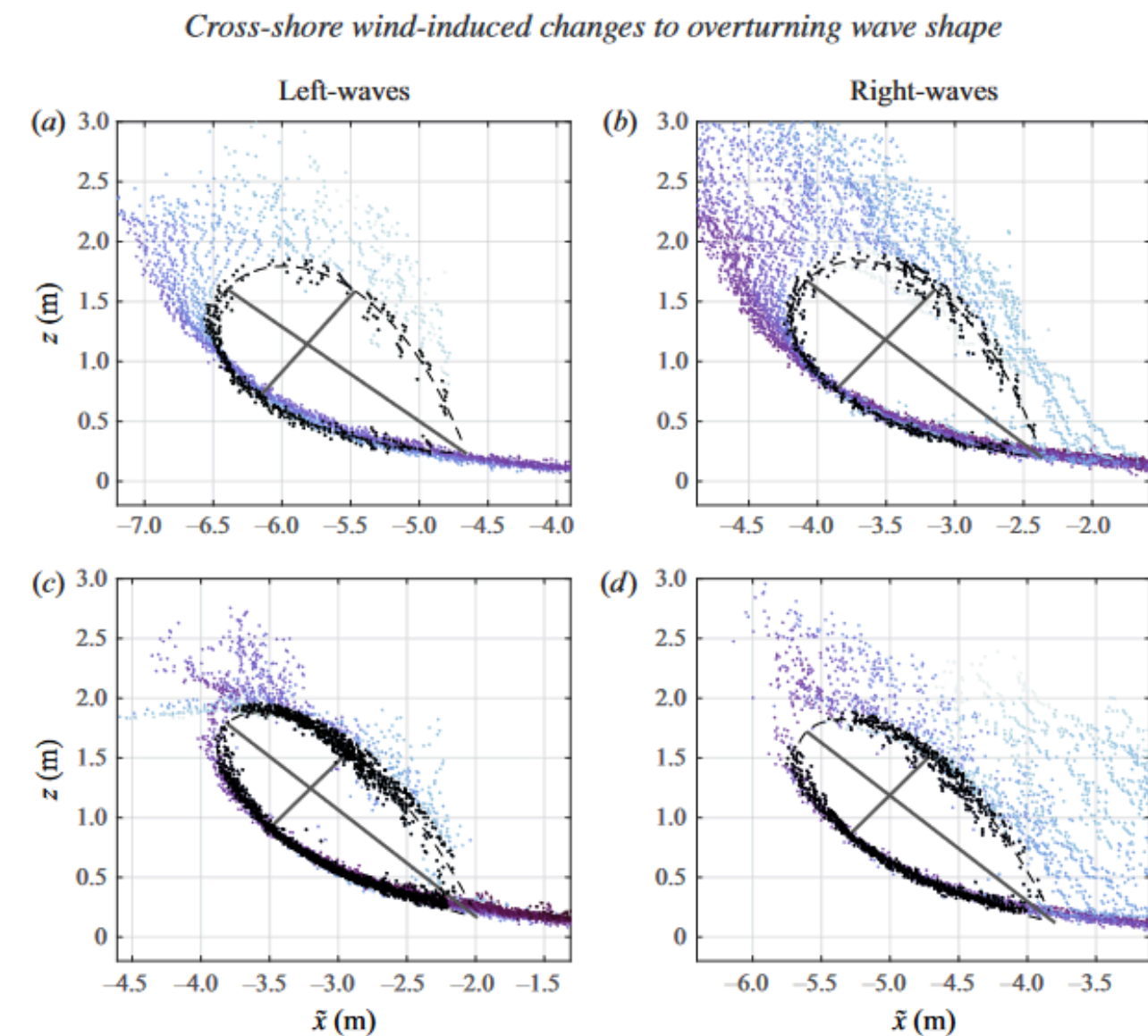
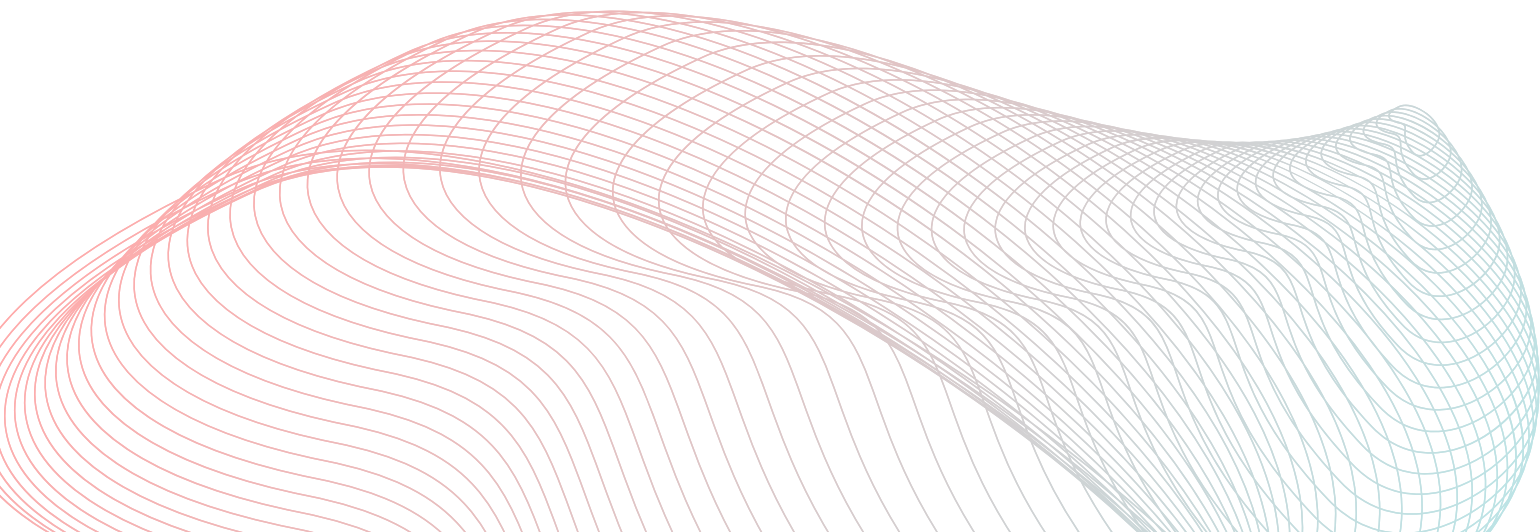
Intern, Summer 2023

Surf Ranch R&D (Wave Systems Engineering)



Cross-Shore Wind-Induced Changes to Wave Shape

- In Conjunction, with Scripps and USC on a continuation of [this project](#).
- Recently, concluded that cross-shore winds does, in fact, affect wave shape
- Now, quantifying how winds affect wave shapes
- Added wind vectors to previous simulations

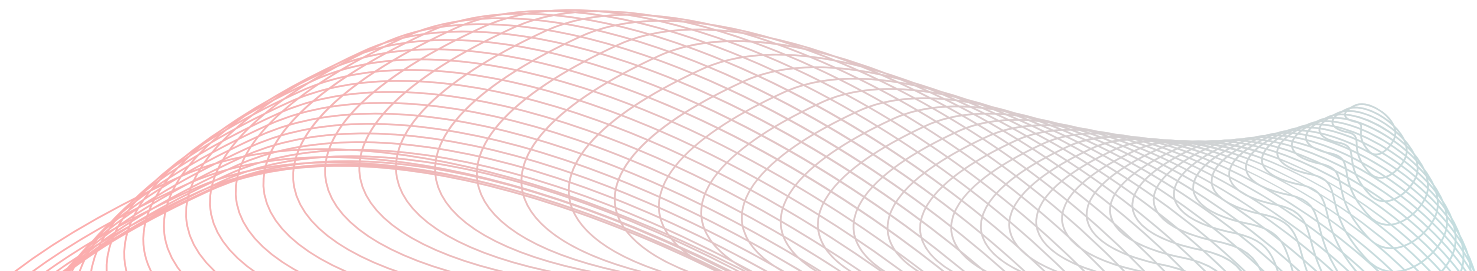


CFD Prototyping and Validation

- Simulated wave shape after strip-light installation
- Quantified effects of strip lights on:
 - Barrel vortex ratio
 - Resultant force on 'Wave-Side' posts
 - Overall wave shape and 'rideability'
- Designed experiment to validate or invalidate CFD results
 - Installed 20 boards to replicate strip-light installation
 - Surveyed surfers to determine affect on overall wave shape and 'rideability'



****CFD Simulation and Results not pictured****



OTHER PROJECTS

Predicting Bee Colony Success with ML

- Analyzed tabular data of environmental variables and several hours of audio files
- Compared 10 Classifier Models from the SciKitLearn Python Package
- Selected most accurate classification model
- Selected best features of the most accurate classification model via Lasso Regularization
- Achieved 85% accuracy on test data set

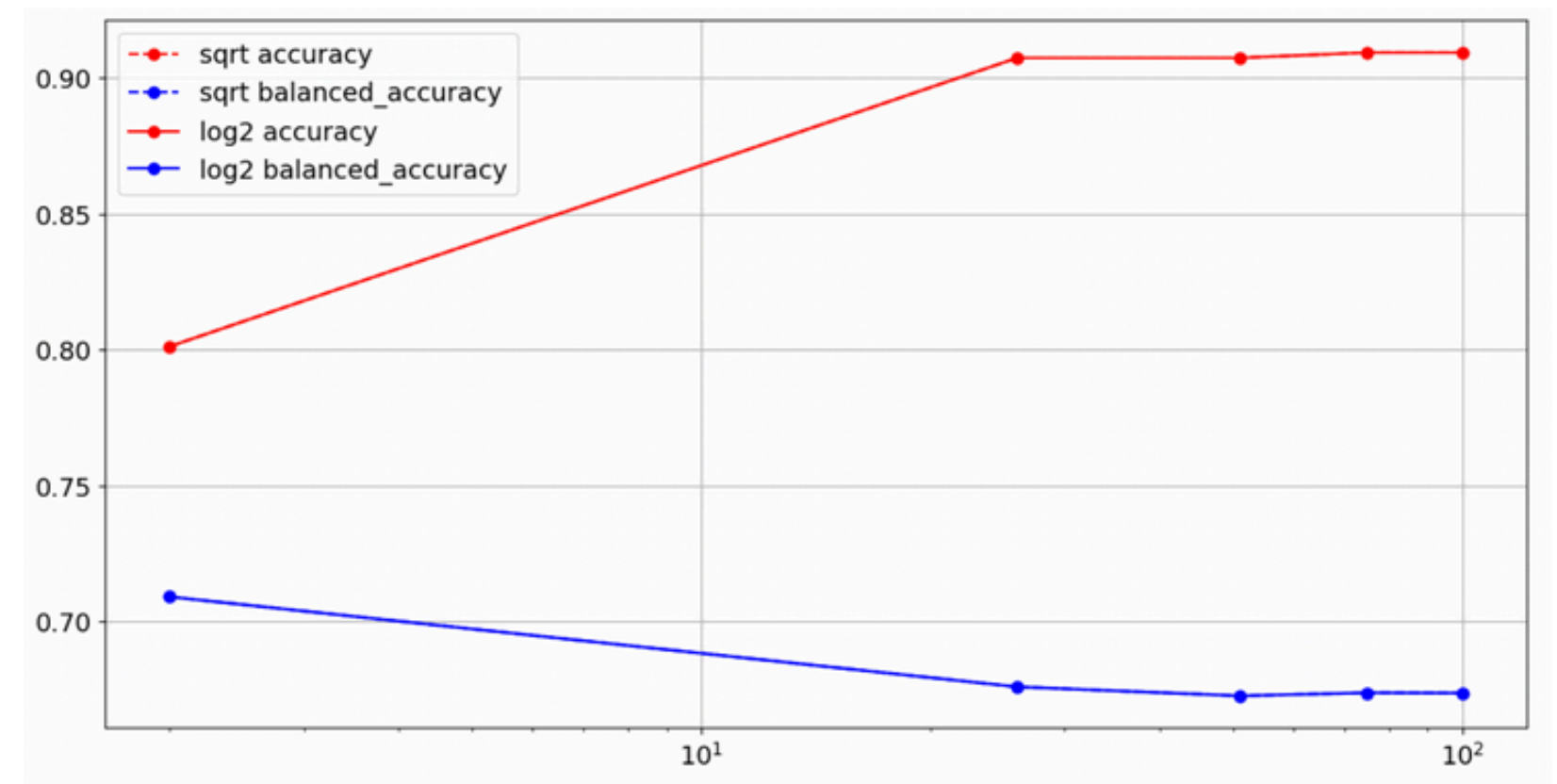
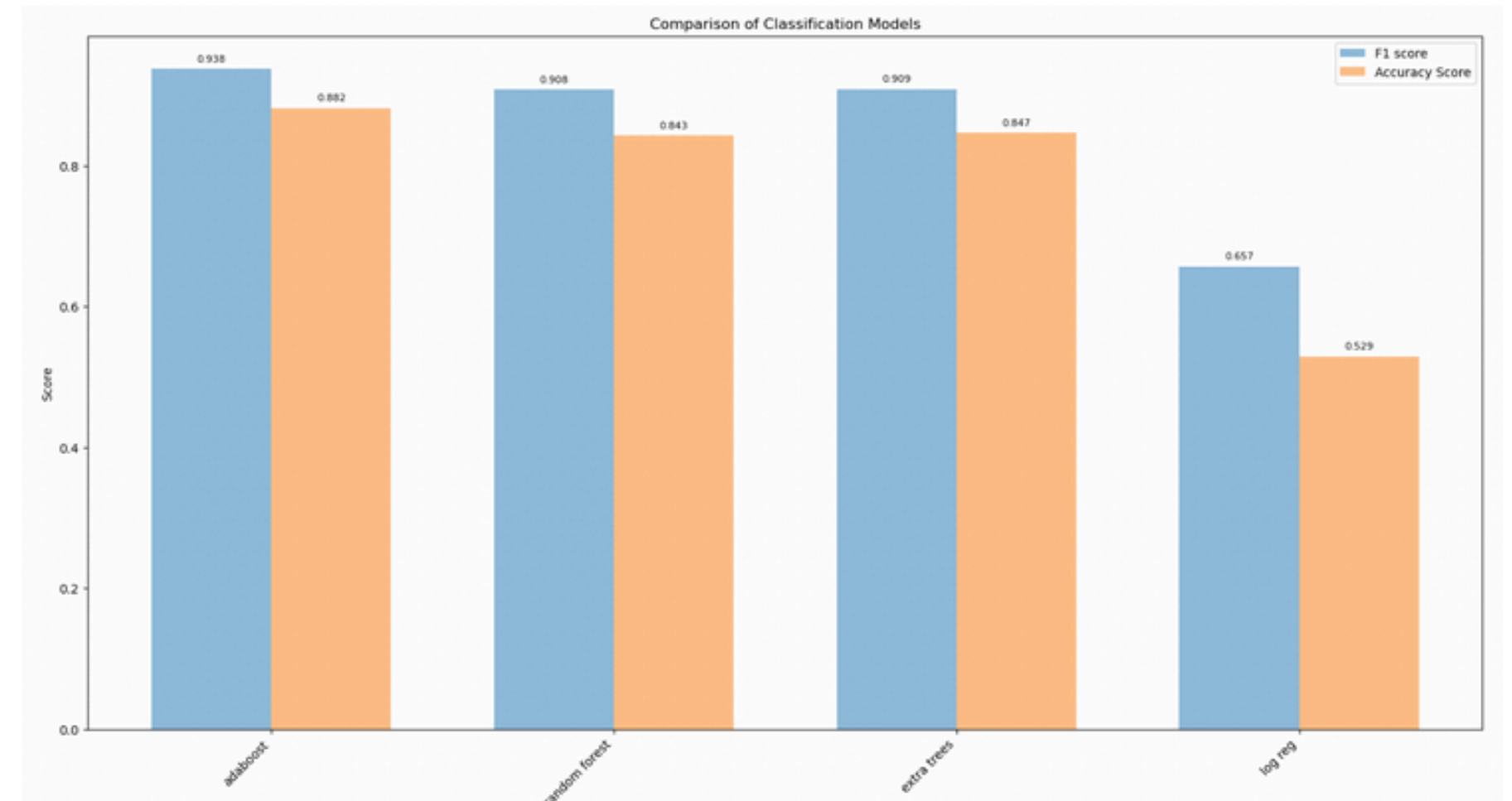
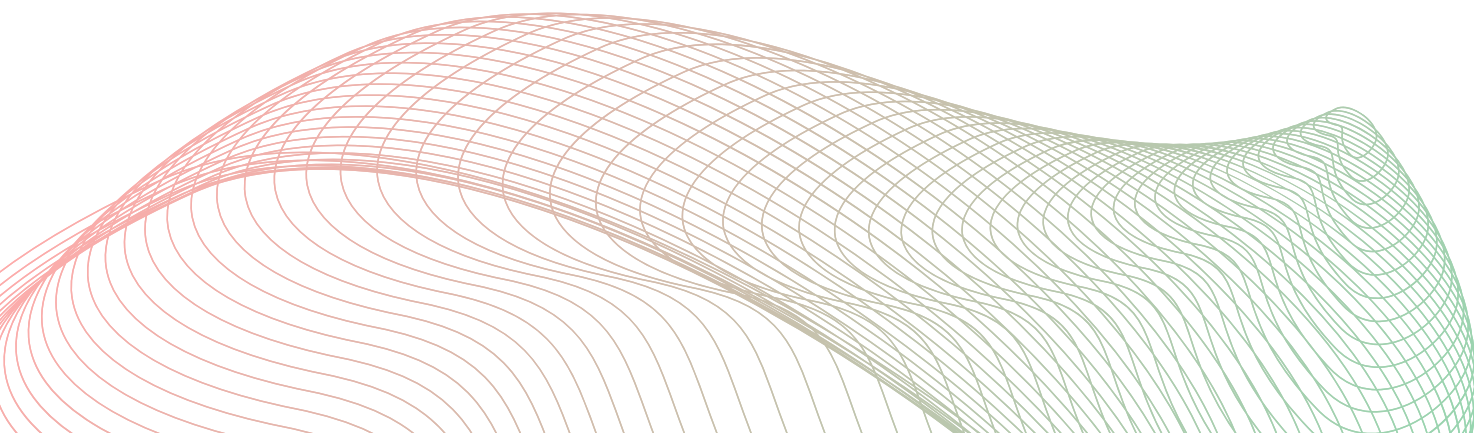
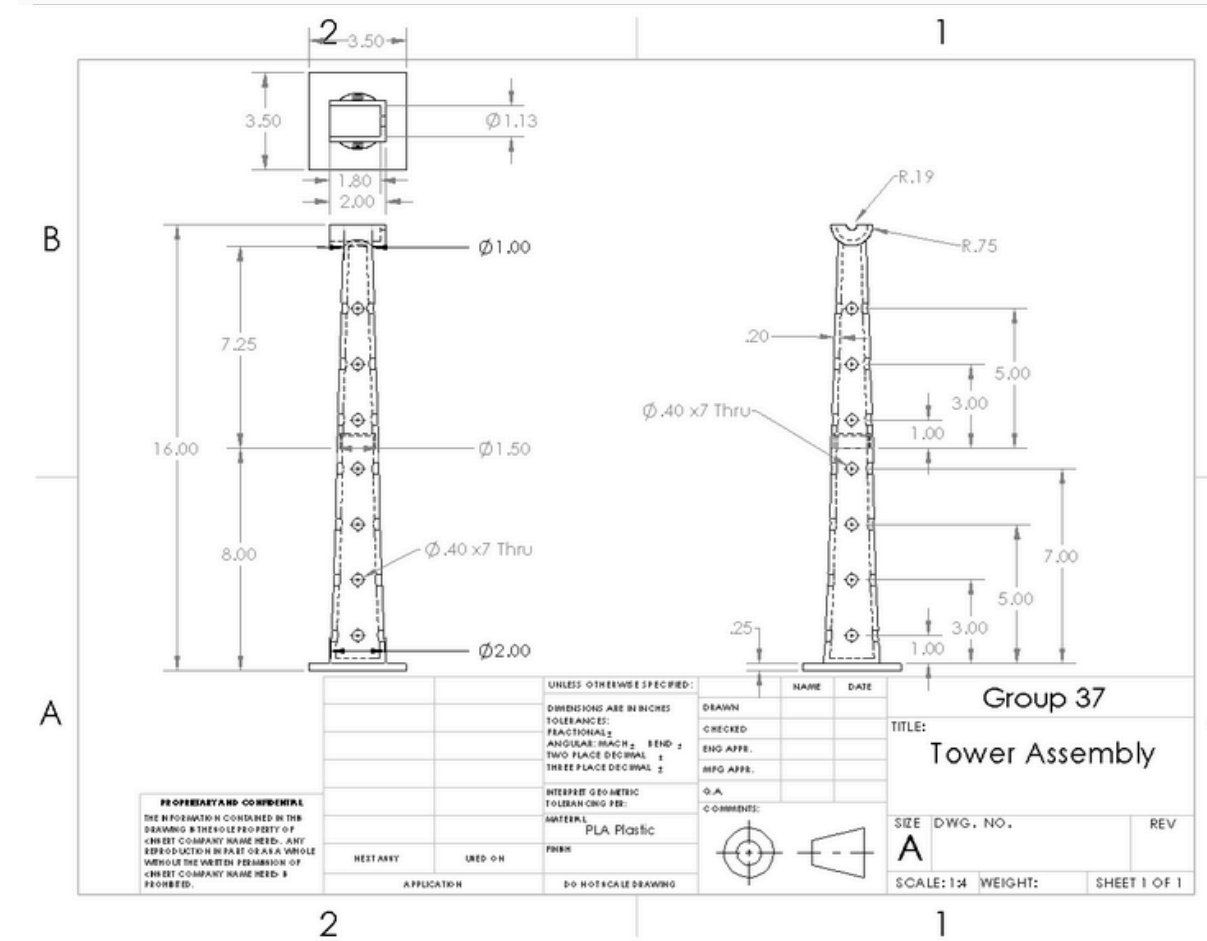
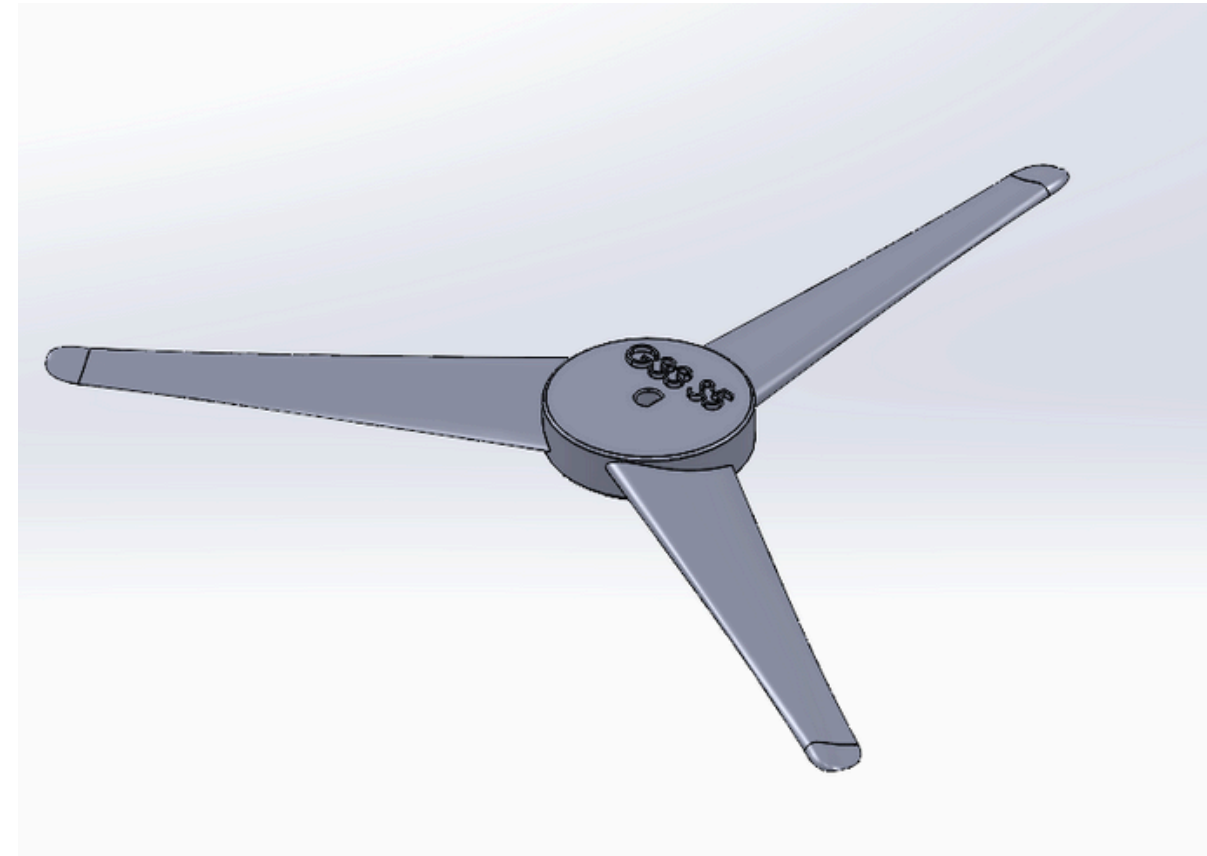


Figure 4: Extra Trees Classifier Accuracies

Wind Turbine

- Led Airfoil Design
- Airfoil Parameters:
 - AoA = 14°
 - AoT = 12°
 - 3 Blades
- Two part tower design for 3D printing
 - Optimized for Stiffness using FEA
 - Holes to stay under weight maximum



Combat Robot

- 3lb Spinner Robot for Intra-School Competition
- Composite Body
 - Aluminum 6061 covered HDPE
- Steel Spinner
- Belt Drive

